Dinesh Arunachalam Robotics Software Engineer

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Summary

Robotics software engineer (AI & Robotics) building perception, embedded and AI-driven systems for real-world robots. Hands-on across sensing, firmware, ROS 2 software and hardware integration, with a focus on physical AI.

Education

B.Tech Computer Science Engineering (Specialization in AI & Robotics) 07/2023 - Expected 05/2027 | Chennai
Vellore Institute of Technology
CGPA: 8.8/10

Professional Experience

Robotics Software Engineer (Intern), *Sentrix Robotics* 02/2026 - 04/2026 | Chennai
(Physical AI startup for robotic dexterity)

- Integrated a motion-capture glove with a robotic arm for precise, real-time teleoperation
- Developed C++ firmware on ESP32-C6 and ROS 2 middleware over serial, mapping 5 glove sensor channels to arm joint commands with low latency.
- Integrated a custom PCB with the glove, interfacing sensors over I2C and UART.
- Set up a ROS 2 Humble arm simulation in NVIDIA Isaac Sim (URDF) for validation.
- Extended the glove into a VR input device, using an attached Meta VR controller for spatial tracking and the glove itself for click inputs.

Projects

Stateful ROS 2 Perception System, *ROS 2, Python, C++, YOLOv8, ByteTrack, OpenCV, SQLite*

- Architected a ROS 2 perception system in C++ and Python that processes real-time RGB video with YOLOv8 and ByteTrack and stores 25+ tracked object instances in a persistent, queryable spatial SQLite database at a time.
- Designed custom ROS 2 service and message interfaces to bridge high-frequency neural network inference with a stateful SQLite backend, supporting multi-object instance tracking and temporal decay.
- Developed a single launch file that configures the camera, tracker and database nodes, plus a natural-language CLI that lets operators query the historical location and spatial context of previously observed objects.

Active Perception for a Mobile Robot, *ROS 2, Python, YOLOv8, OpenCV, VLM | Beetlebot*

- Built a modular ROS 2 system with custom messages on a physical Beetlebot rover (Raspberry Pi, live camera feed) that detects an object and scores the camera view on sharpness, lighting, framing and occlusion, so the robot knows why a view is poor.
- Designed noise, exposure and scale robust quality metrics in OpenCV and combined them into one score that flags the weakest factor.
- Implemented a closed-loop decision layer that picks a corrective move from the weakest factor and re-scores, after a fixed number of attempts the rover returns to the best view seen and stops there.
- Integrated a Gemini vision-language model to judge occlusion and framing, called only after a fast local check passes, which saves API calls.

Skills

- **Languages and Tools:** C++, Python, C, SQL, Linux, Git, Docker, CMake, SQLite, C#, Go
- **Robotics and Computer Vision:** ROS 2, Gazebo, NVIDIA Isaac Sim, MuJoCo, RViz, MoveIt2, Nav2, micro-ROS, OpenCV, YOLO, TensorRT, Vision Language Models, SLAM
- **Embedded Systems and Firmware:** ESP32, Arduino, RaseberryPi, I2C, UART, PWM Control, IMU, Flex Sensors, Encoders, Custom PCB Integration, KiCad, LiDAR
- **AI and Software:** Large Language Models (LLMs), LangChain, LangGraph, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), REST APIs, Flask, Data Structures and Algorithms, Object-Oriented Programming, Robotic Process Automation (RPA) with UiPath
- **Relevant Coursework:** Robot Kinematics and Dynamics, Control Theory, Robotics Perception, Artificial Intelligence, Machine Learning, Linear Algebra and Multivariable Calculus